FIREWALL BASIC THEORY

1. Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.

Steps:

1. Open **Control Panel** and select **Windows Defender Firewall**.
2. Click **Advanced settings**.
3. Select **Outbound Rules → New Rule**.
4. Choose **Program** and click **Next**.
5. Select **This program path** and browse to the Spotify application file (Spotify.exe).
6. Select **Block the connection** and click **Next**.
7. Select **Domain, Private, and Public** profiles.
8. Enter the rule name **Block Spotify Traffic** and click **Finish**.

2. Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).

Steps:

1. Open **Control Panel → Windows Defender Firewall**.
2. Click **Advanced settings → Inbound Rules → New Rule**.
3. Select **Port** and click **Next**.
4. Select **TCP** and enter **443** under Specific local ports.
5. Select **Allow the connection** and click **Next**.
6. Select **Domain, Private, and Public** profiles.
7. Enter the name **Allow Chrome HTTPS Port 443** and click **Finish**.

3. Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall.

Hint: Use the 'Scope' tab when creating a new rule to specify the IP address.

Steps:

1. Open **Control Panel → Windows Defender Firewall**.
2. Click **Advanced settings**.
3. Select **Inbound Rules → New Rule**.
4. Choose **Custom** and click **Next**.

5. Select **All programs** and click **Next**.
6. In the **Scope** tab, under *Remote IP address*, select **These IP addresses** and click **Add**.
7. Enter the IP address 123.45.67.89 and click **OK**.
8. Click **Next**, select **Block the connection**, and click **Next**.
9. Select **Domain, Private, and Public** profiles.
10. Name the rule **Block Specific IP Address** and click **Finish**.

4. Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each.

Hardware Firewall vs Software Firewall:-

Difference	Hardware Firewall	Software Firewall
1. Installation	Installed as a physical device in the network.	Installed on an individual computer.
2. Protection	Protects multiple devices across a network.	Protects a specific device.
3. Cost	Generally more expensive.	Often included with the operating system.

Use Cases:

- **Hardware Firewall:** Used in large offices to protect the entire company network.
- **Software Firewall:** Used on personal computers to control incoming and outgoing network traffic, e.g., Windows Defender Firewall.